

Assignment

TU Stock Exchange – Get Rich (theoretically)

In this Programmierpraktikum, you will build a toy stock exchange application using real-world stock information. This application needs to meet the requirements listed below.

You are free to add anything that you find fun to build if you can defend it to be of any value for the users. Your software project is going to be a distributed system, i.e., have a separate frontend, a corresponding backend, and a database. You have the freedom to choose the technologies you wish to use, if the conditions described below are met.

Throughout the semester, there will be several deliverables, which you are required to prepare and submit in order to pass the course. The deadlines for all exam elements are provided below.

Requirements

1. Users can *buy* and *sell* stocks on the exchange.
2. Trades (i.e., buying and selling) use current real-world stock prices (accurate to the hour).
3. For testing purposes, it should also be possible to use test data for stock prices (e.g., to simulate a market crash, among other things).
4. Users can set up automatic trades based on price thresholds, e.g., *buy* 5 stocks of **GME** for 20 USD/stock or less.
5. Users have some *starting capital* after registration, and compete on a leaderboard based on *net worth*.
6. A user with a net worth that is too low to do anything meaningful on the platform can default (design a good process for this).
7. Users can view the historic performance of the portfolio of any user.

Further Information

1. You are expected to use external APIs (e.g., for using real stock prices).
2. A good free-for-personal-use API for stock prices is provided by *yahoofinance*.¹
3. Come up with a strategy that allows for calling multiple API endpoints at the same time fast and without being shut out by the provider.
4. You have (almost) free choice of your tech stack. You can experiment with technologies you like.
5. Any platform is fine as long as you build a distributed system, i.e., a separate frontend, backend, and a database.
6. Until the final presentation, you should deploy your project on GCP (we will provide free credits for you).

Deliverable 1: 10 Project Diaries

[30 Points]

Each team member submits a summary of max. 0.5 page of the work done in the previous week. You submit the diaries combined in one single PDF file in your GitHub/GitLab repository. These diaries are to be submitted on **Wednesdays (e.o.d)**. The submission dates are **May 6, 2026 to July 8, 2026**.

¹See here for an example use in Python: <https://github.com/ranaroussi/yfinance>

Deliverable 2: *Presentations*

[25 Points]

Over the course of the semester, you need to present the progress on your software project. This includes *one midterm presentation*, *one final presentation* as well as a *poster session* coordinated by the faculty. For each presentation, you need to submit your slides or poster via ISIS. Although we do not encourage the groups to choose more than one speaker to present, we expect all group members to attend the presentations.

(a) Midterm Presentation (10 Points) In the midterm presentation you present the current state of your work. This should showcase your assignment, how you broke down the task into subtasks and milestones, how you organized your group, present your tech stack, describe your system architecture, and the current state of the project etc. The presentation lasts approx. 8min with a 5min Q&A. The slides are due on **June 3, 2026 (Wednesday, e.o.d.)**, the presentations are on **June 4, 2026 (Thursday, 12:15h)**.

(b) Final Presentation (15 Points) In the final presentation you present your product, including a demo (which can also be pre-recorded), drawbacks of your implementations, and an outlook on open to-dos. The presentation lasts approx. 15min with a 5min Q&A. Different to the midterm presentation, you are expected to prepare a poster instead of slides. The poster is due on **July 8, 2026 (Wednesday, e.o.d.)**, the presentations are on **July 9, 2026 (Thursday, 12:15h)**.

(c) Poster Session All Programmierpraktika are expected to attend the official poster session to share the results. We encourage you, however, to join it as this is a good opportunity to see what your peers have achieved. The poster session is on **t.b.d.**.

Deliverable 3: *Source Code + Documentation, Report*

[45 Points]

The submission of your source code with a concise online documentation (30P) is due on **August 5, 2026 (Wednesday, e.o.d.)**. The report (15P) should be max. 5 pages (excl. figures, tables, etc.) and is due on **August 12, 2026 (Wednesday, e.o.d.)** – it should be clear who created each section.

Additional Information

- For group selection please use the ISIS tool “Group Selection” and the respective forum. Each group may contain up to 7 members, with the aim being max. 5 groups of 7 people. Please let us know by **April 29, 2026 (Wednesday, e.o.d.)** how your groups are organized.
- As a group you should designate a person responsible for organization, task delegation, and communication with the module supervisors. This person is the **project manager (PM)** of your group. Project management tools are highly recommended for organization, documentation, and overview of work.
- Your project should be hosted on GitHub/GitLab, and the module supervisors must be added as collaborators to the repository.